

# Applied Physics-II

A complete student textbook for wave motion, optics, electromagnetism and modern physics with formulas, diagrams, animations, solved numerical problems, expected questions, model paper and full answers.

|                                                            |                                            |
|------------------------------------------------------------|--------------------------------------------|
| <div>COURSE CODE</div> 2003                                | <div>COURSE TITLE</div> Applied Physics-II |
| <div>PROGRAMME</div> Diploma in Engineering and Technology | <div>SEMESTER</div> 2                      |
| <div>CREDITS</div> 3                                       | <div>CATEGORY</div> Basic Science          |
| <div>PERIODS / WEEK</div> 3 (L:3, T:0, P:0)                | <div>PERIODS / SEMESTER</div> 45           |

## 45 Periods of physics practice

Learn principle, formula, diagram and numerical method together. Physics becomes easy when concept and calculation are studied as one unit.

- Waves
- Optics
- Electricity
- Magnetism
- Semiconductor
- LASER

### COURSE OVERVIEW

## What Applied Physics-II is about

Applied Physics-II connects fundamental physics with engineering applications. The course begins with wave motion and sound, moves to lenses and optical fiber, then studies electricity, circuits, magnetism and measuring instruments, and ends with semiconductor devices, photoelectric effect, solar cells, LASERs and nanotechnology.

### Engineering applications

Waves explain sound, vibration and ultrasonic testing. Optics explains lenses, microscopes and fiber communication.

You will calculate wave velocity, frequency, lens power, magnification, resistance, meter bridge balance and galvanometer conversion.

Many physics answers become clear with a neat diagram: convex lens, optical fiber, circuit, galvanometer, p-n junction and laser energy levels.

For every topic, follow this order: definition → physical meaning → formula → diagram → solved example → practice problem → revision.

Write given data, formula, substitution, calculation and final unit.

## 4. Revise diagrams

Practise drawing labelled diagrams quickly and neatly.

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COURSE OBJECTIVES AND OUTCOMES

## Course Objectives

1. To provide students with a broad understanding of physical principles of the universe to help them develop critical thinking and quantitative reasoning skills.
2. To help diploma engineers apply the basic concepts of physics to solve broad-based engineering problems.

## Prerequisite

**Basic knowledge in Physics at Secondary School level.**

You should know basic measurement, units, motion, light, electricity and simple formulas.

Secondary school physics അറിയാമെങ്കിൽ ഈ subject-ൽ numerical problems easier ആയിരിക്കും.

## Course Outcomes

| CO          | Outcome                                                                            | Duration | Level         | Student meaning                                                            |
|-------------|------------------------------------------------------------------------------------|----------|---------------|----------------------------------------------------------------------------|
| CO1         | Calculate the characteristics of waves.                                            | 10 h     | Applying      | Find frequency, wavelength, velocity, beat frequency and acoustics basics. |
| CO2         | Compute the power of lens.                                                         | 11 h     | Applying      | Use lens formula, magnification and power of lens.                         |
| CO3         | Convert galvanometer into ammeter and voltmeter.                                   | 11 h     | Applying      | Use electricity, magnetism and resistance ideas to design meter circuits.  |
| CO4         | Explain semiconductor physics, photoelectric effect, LASER action and nanoscience. | 11 h     | Understanding | Understand diodes, transistors, solar cells and lasers.                    |
| Series Test | Series Test                                                                        | 2 h      | -             | Series Test I after Module 2; Series Test II after Module 4.               |

## CO-PO Mapping Summary

### CO1 → PO1

3, strongly mapped.

### CO2 → PO1

3, strongly mapped.  
Course 2003

CO3 → PO1

3, strongly mapped.

CO4 → PO1

2, moderately mapped.

Mapping scale: 3 - Strongly mapped, 2 - Moderately mapped, 1 - Weakly mapped.

MODULE-WISE LEARNING ROADMAP

45-period plan

| Module                              | Duration              | Main topics                                                                     | Numerical focus                                        | Diagram focus                                       |
|-------------------------------------|-----------------------|---------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------|
| 1. Wave Motion and Its Applications | 10 h                  | SHM, waves, beats, ultrasonics, acoustics                                       | $v=f\lambda$ , $T=1/f$ , beat frequency                | SHM projection, transverse/longitudinal waves       |
| 2. Optics                           | 11 h + Series Test I  | Reflection, refraction, lenses, optical instruments, TIR, fiber                 | Lens formula, power, magnification, critical angle     | Convex lens, TIR, optical fiber                     |
| 3. Electromagnetism                 | 11 h                  | Charge, field, current, resistance, Kirchhoff, bridge, magnetic effects, meters | Ohm's law, resistance, meter bridge, ammeter/voltmeter | Circuit, bridge, galvanometer, meter conversion     |
| 4. Modern Physics                   | 11 h + Series Test II | Semiconductors, photoelectric effect, solar cell, lasers, nanoscience           | Photon energy if required                              | Energy bands, p-n junction, diode V-I, laser levels |

MODULE 1 • CO1

Wave Motion and Its Applications

Official focus: periodic motion, SHM, projection of uniform circular motion, wave characteristics, superposition, beats, ultrasonic waves, numerical problems and acoustics of buildings.

## Module Outcomes

- M1.01 Discuss SHM and properties.
- M1.02 Calculate wave characteristics.
- M1.03 Define ultrasonic waves and list applications.
- M1.04 Understand acoustics of buildings.

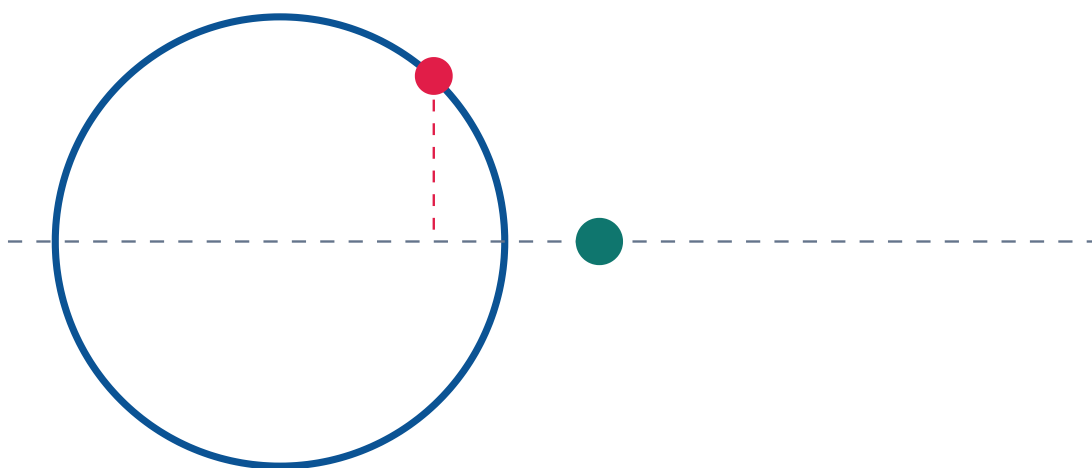
## Periodic motion and SHM

Periodic motion repeats after equal intervals of time. Simple Harmonic Motion is a special periodic motion in which acceleration is directly proportional to displacement and always directed towards mean position.

ഒരേ സമയം കഴിഞ്ഞ് motion വീണ്ടും ആവർത്തിച്ചാൽ periodic motion. SHM-ൽ acceleration എപ്പോഴും mean position-ലേക്ക് ആയിരിക്കും.

## SHM as projection of circular motion

If a particle moves uniformly in a circle, the projection of the particle on any diameter performs SHM. This explains displacement, velocity and acceleration equations.



**Projection on diameter performs SHM**

## SHM formulas

$$x = A \sin(\omega t)$$

$$v = A\omega \cos(\omega t)$$

$$a = -\omega^2 x$$

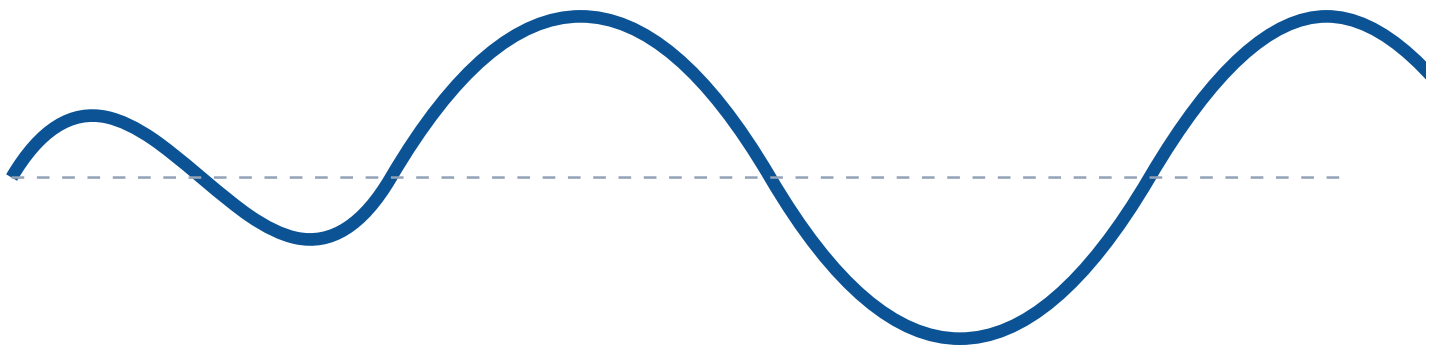
$$T = 1/f, \omega = 2\pi f = 2\pi/T$$

## Wave motion basics

A wave transfers energy from one place to another without transferring matter as a whole. In mechanical waves, particles vibrate about mean positions.

### Transverse wave

Particles vibrate perpendicular to direction of propagation. Examples: string wave, light wave.



**Transverse wave**

### Longitudinal wave

Particles vibrate parallel to direction of propagation. Example: sound wave in air.



## Compressions and rarefactions

## Wave characteristics

| Quantity             | Meaning                               | Unit          |
|----------------------|---------------------------------------|---------------|
| Amplitude A          | Maximum displacement                  | m             |
| Wavelength $\lambda$ | Distance between points in same phase | m             |
| Frequency f          | Oscillations per second               | Hz            |
| Time period T        | Time for one oscillation              | s             |
| Wave velocity v      | Distance travelled per second         | m/s           |
| Phase                | State of vibration                    | radian/degree |

$$v = f\lambda$$

## Superposition and beats

When waves meet, resultant displacement is algebraic sum of displacements. Beats are periodic loud-soft variations produced by two waves of nearly equal frequency.

$$\text{Beat frequency} = |f_1 - f_2|$$

Frequency difference ചെറിയതായാൽ sound loud-soft ആയി കേൾക്കും. അതാണ് beats.

## Ultrasonic waves and acoustics

Ultrasonic waves have frequency above 20 kHz. They are used in crack detection, thickness measurement, ultrasonic cleaning, medical scanning and welding. Acoustics of buildings studies echo, reverberation, noise and methods to improve sound clarity in halls.

## Echo



## Reverberation

Persistence of sound due to multiple reflections.

## Control

Use curtains, carpets, acoustic panels, false ceiling and avoid flat reflecting surfaces.

## Solved examples

### Wave velocity

$$f = 500 \text{ Hz}, \lambda = 0.68 \text{ m}. v = f\lambda = 500 \times 0.68 = 340 \text{ m/s}.$$

### Beat frequency

$$f_1 = 256 \text{ Hz}, f_2 = 260 \text{ Hz}. \text{ Beat frequency} = |260 - 256| = 4 \text{ Hz}.$$

## MODULE 2 • CO2

# Optics

Official focus: reflection, refraction, refractive index, lenses, convex lens image formation, lens formula, power, magnification, aberrations, microscope, telescope, TIR and optical fiber.

## Module Outcomes

- M2.01 Explain laws of optics and convex lens image formation.
- M2.02 Apply distance relation and discuss lens defects.
- M2.03 Explain optical instruments.
- M2.04 Describe TIR and optical fiber.

## Reflection and refraction

Reflection is bouncing back of light. Refraction is bending of light when it passes between media due to speed change.

Light ഒരു medium-ൽ നിന്ന് മറ്റൊന്നിലേക്ക് പോകുമ്പോൾ speed മാറുന്നതിനാൽ direction മാറും. അതാണ് refraction.

## Laws and refractive index

### Reflection

1. Incident ray, reflected ray and normal lie in same plane.
2. Angle of incidence equals angle of reflection.

### Refraction

1. Incident ray, refracted ray and normal lie in same plane.
2.  $\sin i / \sin r$  is constant.

$$n = c/v$$

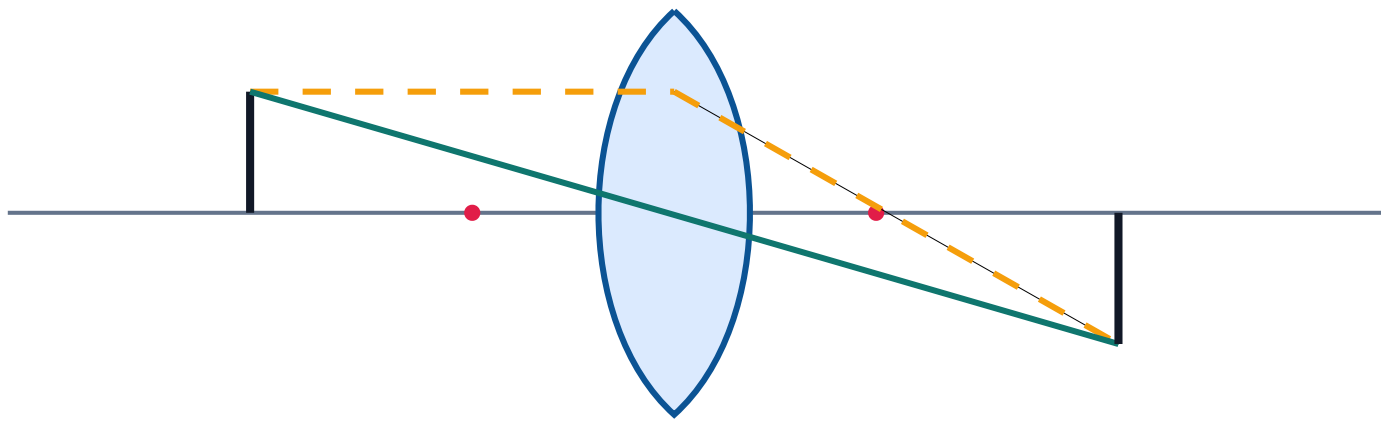
## Lenses and formulas

### Convex lens

Thicker at centre, converges rays, can form real or virtual images.

### Concave lens

Thinner at centre, diverges rays and generally gives virtual erect diminished image.



**Convex lens image formation**

Lens formula:  $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

Power:  $P = \frac{1}{f}$ , where  $f$  is in metre. Unit: dioptre.

Magnification:  $m = \frac{v}{u} = \frac{\text{image height}}{\text{object height}}$

**Exam Tip:** Convert focal length from centimetre to metre before calculating power.

## Lens defects and instruments

### Chromatic aberration

Different colours focus at different points, causing coloured fringes.

### Spherical aberration

Rays through different parts of spherical lens do not meet at one focus.

### Simple microscope

Single convex lens of short focal length producing magnified virtual image.

### Astronomical telescope

## Total internal reflection and optical fiber

TIR occurs when light travels from denser to rarer medium with angle of incidence greater than critical angle. Optical fiber guides light by repeated TIR through core.

$$\sin C = 1/n$$

### Light guided by total internal reflection



## Solved examples

### Power of lens

$$f = 25 \text{ cm} = 0.25 \text{ m. } P = 1/f = 1/0.25 = +4 \text{ D.}$$

### Lens formula

$$u = -30 \text{ cm, } f = +20 \text{ cm. } 1/f = 1/v - 1/u. \quad 1/20 = 1/v + 1/30. \quad 1/v = 1/20 - 1/30 = 1/60. \quad v = 60 \text{ cm.}$$

## MODULE 3 • CO3

# Electromagnetism

Official focus: Coulomb law, electric field, potential, capacitance, current, Ohm law, resistance, conductance, resistor combinations, colour coding, Kirchhoff laws, bridges, magnetic field, induction, Lorentz force, moving coil galvanometer and conversion.

## Module Outcomes

- M3.01 Explain Coulomb law, electric field and potential.
- M3.02 Apply Ohm law and resistance combinations.
- M3.03 Use Kirchhoff laws and meter bridge.
- M3.04 Explain magnetic effect and galvanometer conversion.

## Electric charge and field

Charge is a basic property causing electric force. Electric field is the region around a charge where another charge experiences force.

Charge ചുറ്റുമുള്ള സ്ഥലത്ത് മറ്റൊരു charge force അനുഭവിക്കും. ആ region ആണ് electric field.

## Coulomb law

$$F = (1/4\pi\epsilon_0) q_1q_2/r^2$$

## Field, potential, capacitance

$$E = F/q, V = W/q, C = Q/V$$

## Ohm law and resistance

Electric current is rate of flow of charge. DC flows in one direction; AC changes direction periodically.

$$I = Q/t, V = IR$$

$$R = \rho l/A, G = 1/R$$

Wire നീളം കൂടുമ്പോൾ resistance കൂടും. Area കൂടുമ്പോൾ resistance കുറയും.

## Series

$$R_s = R_1 + R_2 + R_3 + \dots$$

## Parallel

$$1/R_p = 1/R_1 + 1/R_2 + \dots$$

## Colour coding

Digits: Black 0, Brown 1, Red 2, Orange 3, Yellow 4, Green 5, Blue 6, Violet 7, Grey 8, White 9. First two bands give digits, third gives multiplier, fourth gives tolerance.

## Kirchhoff, bridge and meter bridge

### KCL

Total current entering a junction equals current leaving.

$$\Sigma I = 0$$

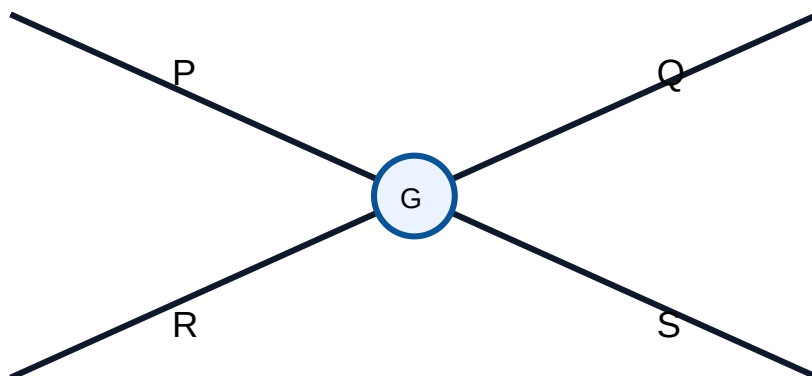
### KVL

Algebraic sum of potential differences in a closed loop is zero.

$$\Sigma V = 0$$

Wheatstone balance:  $P/Q = R/S$

Meter bridge:  $R/S = l/(100-l)$



**Wheatstone bridge**

# Magnetism and galvanometer

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Faraday law:  $\epsilon = -N \frac{d\Phi}{dt}$

Lorentz force:  $F = qvB \sin\theta$

Current conductor:  $F = BIL \sin\theta$

A moving coil galvanometer detects small current. It becomes an ammeter by a low resistance shunt in parallel and a voltmeter by a high resistance in series.

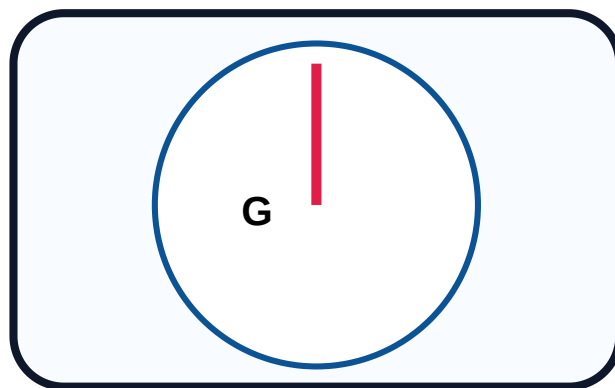
Ammeter ആക്കാൻ parallel shunt, Voltmeter ആക്കാൻ series high resistance.

## Ammeter conversion

$$S = I_g G / (I - I_g)$$

## Voltmeter conversion

$$R = V / I_g - G$$



Moving coil galvanometer

## Solved examples

### Ohm law

$$R = 6 \, \Omega, I = 2 \, \text{A}. V = IR = 2 \times 6 = 12 \, \text{V}.$$

Parallel resistance

6 Ω and 3 Ω in parallel:  $1/R = 1/6 + 1/3 = 3/6 = 1/2$ .  $R = 2\ \Omega$ .

Galvanometer to ammeter

$G = 99\ \Omega$ ,  $I_g = 0.01\text{ A}$ ,  $I = 1\text{ A}$ .  $S = I_g G / (I - I_g) = (0.01 \times 99) / (1 - 0.01) = 0.99 / 0.99 = 1\ \Omega$ . Connect 1 Ω shunt in parallel.

MODULE 4 • CO4

Modern Physics

Official focus: semiconductor physics, energy bands, intrinsic/extrinsic semiconductors, p-n junction, diode V-I characteristics, transistors, photoelectric effect, photocells, solar cells, lasers, He-Ne and semiconductor laser, nanoscience and nanotechnology.

Module Outcomes

- M4.01 Discuss diodes and transistors.
- M4.02 Explain photoelectric effect and applications.
- M4.03 Explain LASER action, semiconductor laser and He-Ne laser.
- M4.04 Summarize nanoscience.

Energy bands in solids

In solids, electrons occupy energy bands. Conductors have overlapping bands, insulators have large forbidden gap, and semiconductors have small forbidden gap.

Semiconductor-ൽ forbidden gap ചെറിയതാണ്. അതിനാൽ doping/temperature കൊണ്ട് conductivity മാറ്റം.

Conductor

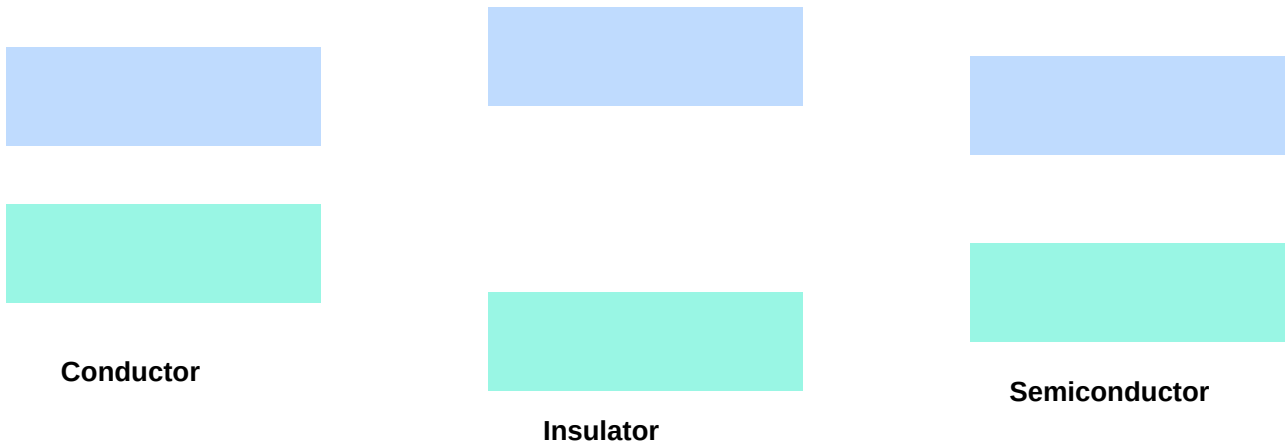
No or very small band gap; many free electrons.

Semiconductor



## Insulator

Large band gap; very low conductivity.



## Semiconductors and p-n junction

### Intrinsic

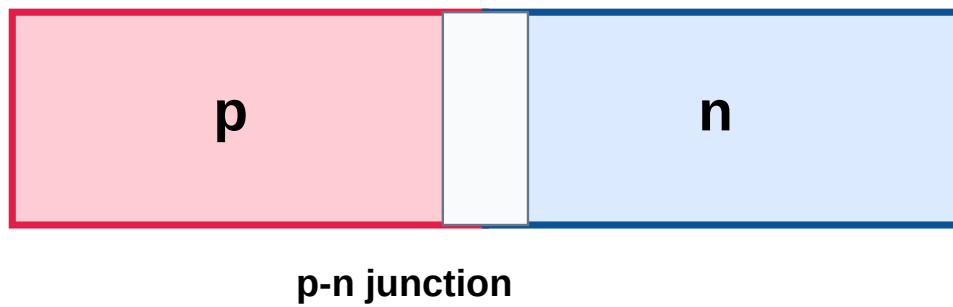
Pure semiconductor with equal electrons and holes.

### Extrinsic

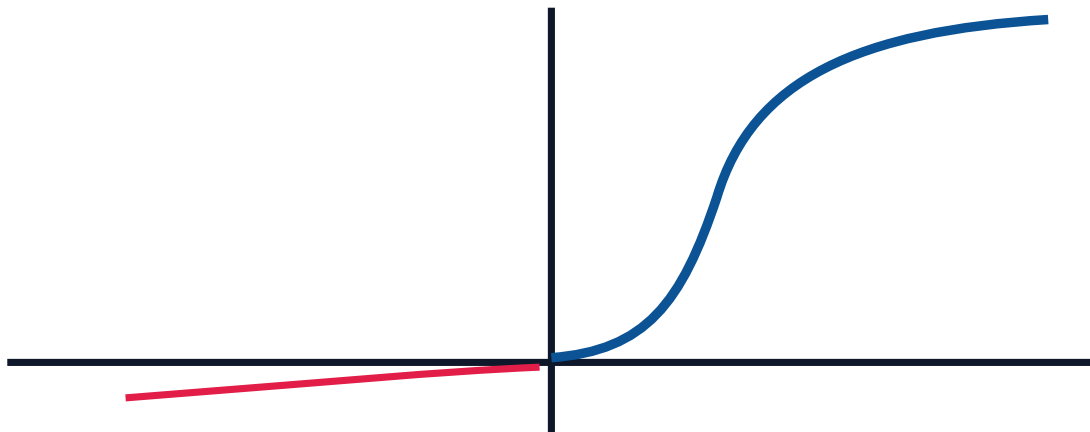
Doped semiconductor: n-type has electrons; p-type has holes as majority carriers.

A p-n junction is formed by joining p-type and n-type semiconductors. Forward bias allows current; reverse bias allows only small leakage current.

Forward bias-ൽ diode current flow ചെയ്യും. Reverse bias-ൽ current വളരെ കുറവ്.



**Diode V-I characteristic**



## Photoelectric effect, solar cell and LASER

Photoelectric effect is emission of electrons when light of suitable frequency falls on a metal. A solar cell converts light energy into electrical energy. LASER means Light Amplification by Stimulated Emission of Radiation.

Photon energy:  $E = hf$

Photoelectric equation:  $hf = \phi + KE_{\max}$

$1 \text{ nm} = 10^{-9} \text{ m}$

Laser light സാധാരണ light പോലെ ചിതറില്ല. അത് coherent, directional, intense ആണ്.



Laser energy levels

### He-Ne laser

Gas laser using helium-neon mixture; common in labs and alignment.

### Semiconductor laser

Compact laser diode used in optical communication and devices.

### Nanoscience

Study of materials at nanometre scale; properties change due to size.

### Applications

Laser cutting, welding, surgery, optical communication; nanotech in sensors, medicine, coatings and energy storage.

FORMULA BANK

# Important formulas module-wise

Course 2008

## SHM and waves

$$x = A \sin(\omega t)$$

$$v = A\omega \cos(\omega t)$$

$$a = -\omega^2 x$$

$$T = 1/f$$

$$\omega = 2\pi f = 2\pi/T$$

$$v = f\lambda$$

$$\text{Beat frequency} = |f_1 - f_2|$$

## Optics

$$n = c/v$$

$$1/f = 1/v - 1/u$$

$$P = 1/f, f \text{ in metre}$$

$$m = v/u = h_i/h_o$$

$$\sin C = 1/n$$

## Electricity

$$F = (1/4\pi\epsilon_0) q_1 q_2 / r^2$$

$$E = F/q$$

$$V = W/q$$

$$C = Q/V$$

$$V = IR$$

$$R = \rho l/A$$

$$G = 1/R$$

$$R_s = R_1 + R_2 + \dots$$

$$1/R_p = 1/R_1 + 1/R_2 + \dots$$

## Bridge, magnetism, meters

$$\text{KCL: } \Sigma I = 0$$

$$\text{KVL: } \Sigma V = 0$$

$$P/Q = R/S$$

$$R/S = l/(100-l)$$

$$\varepsilon = -N d\Phi/dt$$

$$F = qvB \sin\theta$$

$$F = BIL \sin\theta$$

$$S = I_g G / (I - I_g)$$

$$R = V/I_g - G$$

# Modern Physics

$$E = hf$$

$$hf = \phi + KE_{\max}$$

$$1 \text{ nm} = 10^{-9} \text{ m}$$

## UNITS AND SYMBOLS BANK

# Symbols and SI units

Course 2009

| Quantity         | Symbol    | Unit        |
|------------------|-----------|-------------|
| Displacement     | x         | m           |
| Amplitude        | A         | m           |
| Frequency        | f         | Hz          |
| Time period      | T         | s           |
| Wave velocity    | v         | m/s         |
| Wavelength       | $\lambda$ | m           |
| Refractive index | n         | No unit     |
| Focal length     | f         | m           |
| Power of lens    | P         | D           |
| Current          | I         | A           |
| Voltage          | V         | V           |
| Resistance       | R         | $\Omega$    |
| Resistivity      | $\rho$    | $\Omega$ m  |
| Conductance      | G         | S           |
| Capacitance      | C         | F           |
| Charge           | Q         | C           |
| Electric field   | E         | N/C         |
| Magnetic field   | B         | T           |
| Energy           | E         | J           |
| Nanometre        | nm        | $10^{-9}$ m |

## IMPORTANT DIAGRAMS BANK

# Diagrams to practise

## Module 1

- SHM projection
- Transverse wave
- Longitudinal wave
- Beat concept

## Module 2

- Convex lens ray diagram
- Total internal reflection
- Optical fiber propagation
- Microscope/telescope

## Module 3

- Ohm law circuit
- Series and parallel circuits
- Wheatstone bridge
- Meter bridge
- Moving coil galvanometer
- Ammeter/voltmeter conversion

## Module 4

- Energy band diagrams
- p-n junction
- Diode V-I characteristics
- pnp/npn symbols
- Photoelectric setup
- Solar cell
- Laser energy levels

## Diagram method

1. Draw large enough.
2. Use neat labels.
3. Show direction of rays/current where needed.



4. Write one explanation below.

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### SOLVED NUMERICAL EXAMPLES

## Calculation practice from all numerical modules

### Time period and frequency

50 oscillations in 10 s.  $f = 50/10 = 5 \text{ Hz}$ .  $T = 1/f = 1/5 = 0.2 \text{ s}$ .

### Critical angle

$n = 1.5$ .  $\sin C = 1/n = 1/1.5 = 0.6667$ .  $C \approx 41.8^\circ$ .

### Law of resistance

$l=2 \text{ m}$ ,  $A=1 \times 10^{-6} \text{ m}^2$ ,  $\rho=1.7 \times 10^{-8} \Omega\text{m}$ .  $R=\rho l/A=(1.7 \times 10^{-8} \times 2)/(1 \times 10^{-6})=0.034 \Omega$ .

### Meter bridge

Balance length  $l=40 \text{ cm}$ ,  $S=6 \Omega$ .  $R/S=l/(100-l)$ .  $R/6=40/60$ .  $R=4 \Omega$ .

### Galvanometer to voltmeter

$G=50 \Omega$ ,  $I_g=5 \text{ mA}=0.005 \text{ A}$ ,  $V=10 \text{ V}$ .  $R=V/I_g-G=10/0.005-50=2000-50=1950 \Omega$ .

### EXPECTED QUESTIONS

## Module 1

### Very short

1. Define SHM.
2. What is wavelength?
3. Define ultrasonic waves.

### Numerical

1. Find wave velocity.
2. Find beat frequency.

### Long

1. Explain SHM projection.
2. Explain acoustics defects and remedies.

## Module 2

### Very short

1. Define refractive index.
2. What is lens power?
3. Define critical angle.

### Numerical

1. Find power of lens.
2. Solve lens formula problem.

### Long

1. Explain convex lens image formation.
2. Explain optical fiber.

## Module 3

### Very short

1. State Ohm law.
2. Define capacitance.
3. What is KCL?

### Numerical

1. Ohm law.

2. Resistance combination.
3. Galvanometer conversion.

### Long

1. Explain moving coil galvanometer.
2. Explain meter bridge.

## Module 4

### Very short

1. Define semiconductor.
2. What is population inversion?
3. What is 1 nanometre?

### Diagram

1. Energy band diagram.
2. Diode V-I curve.
3. Laser energy levels.

### Long

1. Explain p-n junction diode.
2. Explain laser action.

## Objective / MCQ

| No | Question                        | Options                                                      |
|----|---------------------------------|--------------------------------------------------------------|
| 1  | Wave relation is                | A $v=f/\lambda$ B $v=f\lambda$ C $f=v\lambda$ D $\lambda=vf$ |
| 2  | Unit of lens power is           | A watt B dioptre C tesla D ohm                               |
| 3  | Series equivalent resistance is | A less than least B product C sum D zero                     |
| 4  | Majority carriers in n-type are | A holes B electrons C photons D protons                      |
| 5  | LASER light is                  | A incoherent B divergent C coherent D white                  |

### PRACTICE PROBLEMS

## Module-wise practice with answers

### Module 1 numericals

2.  $T=0.02$  s. Find  $f$ . **Ans: 50 Hz**  
3. 512 Hz and 520 Hz. Beat? **Ans: 8 Hz**

## Module 2 numericals

- Find  $P$  for  $f=20$  cm. **Ans: +5 D**
- Find  $f$  for  $P=-2$  D. **Ans: -0.5 m**
- Find  $C$  for  $n=1.414$ . **Ans:  $45^\circ$**

## Module 3 numericals

- $I=3$  A,  $R=4\ \Omega$ . Find  $V$ . **Ans: 12 V**
- $2\ \Omega, 3\ \Omega, 5\ \Omega$  in series. **Ans: 10  $\Omega$**
- $4\ \Omega$  and  $4\ \Omega$  in parallel. **Ans: 2  $\Omega$**

## Module 4 practice

- Draw energy bands.
- Explain forward and reverse bias.
- Write four LASER applications.

## Self-assessment

- ✓ I can write formulas with units.
- ✓ I can solve lens power problems.
- ✓ I can solve resistance problems.
- ✓ I can draw optical fiber diagram.
- ✓ I can explain laser action.

### QUICK REVISION

# Last-day revision sheet

## Module 1

- $a = -\omega^2 x$
- $v = f\lambda$
- $\text{Beat} = |f_1 - f_2|$
- Reverberation is persistence of sound.

## Module 2

- $n = c/v$
- $1/f = 1/v - 1/u$
- $P = 1/f$  metre
- TIR: denser to rarer and  $i > C$ .

## Module 3

- $V = IR$
- $R = \rho l/A$
- KCL and KVL
- Ammeter: shunt, voltmeter: series R

## Module 4

- p-n junction
- Photoelectric effect
- Laser needs population inversion
- $1 \text{ nm} = 10^{-9} \text{ m}$

**Common Mistakes:** using cm instead of metre for lens power, forgetting units, mixing ammeter and voltmeter conversion, and drawing unlabeled diagrams.

MODEL QUESTION PAPER WITH FULL ANSWERS

## APPLIED PHYSICS-II • Maximum Marks: 75 • Time: 3 Hours

Fresh model paper based on the syllabus. Pattern: very short, short/numerical, and long/application questions.

## PART A - Answer all. [10 × 1 = 10]

1. Define SHM.
2. Write  $v$ ,  $f$ ,  $\lambda$  relation.
3. What is reverberation?
4. Define power of lens.
5. State one TIR condition.
6. State Ohm's law.
7. SI unit of capacitance?
8. What is connected to convert galvanometer into ammeter?
9. What is population inversion?
10. Write one nanometre in metre.

## PART B - Answer any five. [5 × 5 = 25]

1. Explain SHM as projection of circular motion.
2. A wave has  $f=250$  Hz and  $\lambda=1.36$  m. Calculate  $v$ .
3. Explain chromatic and spherical aberration.
4. A convex lens has  $f=25$  cm. Find power.
5. State Kirchhoff laws and one application.
6. Find equivalent resistance of  $4\ \Omega$  and  $6\ \Omega$  in series and parallel.
7. Explain photoelectric effect and one application.

## PART C - Answer any four. [4 × 10 = 40]

1. Explain transverse and longitudinal waves; derive  $v=f\lambda$  and solve:  $f=680$  Hz,  $v=340$  m/s. Find  $\lambda$ .
2. Explain TIR and optical fiber with diagram and applications.
3. Describe moving coil galvanometer and conversion into ammeter and voltmeter.
4.  $G=99\ \Omega$ ,  $I_g=0.01$  A. Find shunt for 1 A ammeter and explain principle.
5. Explain energy bands and compare conductor, semiconductor and insulator.
6. Explain LASER action with energy level diagram, characteristics and applications.

## ANSWER KEY / COMPLETE SOLUTIONS

## Part A Answers

| No | Answer                                                                                        |
|----|-----------------------------------------------------------------------------------------------|
| 1  | Periodic motion with acceleration proportional to displacement and directed to mean position. |
| 2  | $v=f\lambda$ .                                                                                |
| 3  | Persistence of sound due to multiple reflections after source stops.                          |
| 4  | Reciprocal of focal length in metre: $P=1/f$ .                                                |
| 5  | Light travels denser to rarer medium and $i>C$ .                                              |
| 6  | At constant temperature, $V$ is proportional to $I$ ; $V=IR$ .                                |
| 7  | Farad.                                                                                        |
| 8  | Low resistance shunt in parallel.                                                             |
| 9  | More atoms in excited state than lower state.                                                 |
| 10 | $10^{-9}$ m.                                                                                  |

## Part B Solutions

### B1 SHM projection

A particle moving uniformly in a circle has a projection on a diameter. This projection moves to and fro along the diameter and performs SHM. If  $A$  is radius and  $\omega$  is angular velocity, displacement  $x=A \sin \omega t$ . Acceleration is directed towards mean position. Malayalam: Circular motion-ന്റെ projection നോക്കുമ്പോൾ oscillation പോലെ കാണുന്നതാണ് SHM.

### B2 Wave velocity

$f=250$  Hz,  $\lambda=1.36$  m.  $v=f\lambda=250 \times 1.36=340$  m/s.

### B3 Lens defects

Chromatic aberration: different colours focus at different points, producing coloured fringes. Spherical aberration: rays through different parts of a spherical lens do not meet at one focus, producing blur.

## B4 Power

$f=25\text{ cm}=0.25\text{ m}$ .  $P=1/f=1/0.25=+4\text{ D}$ .

## B5 Kirchhoff laws

KCL: Total current entering a junction equals total current leaving. KVL: Algebraic sum of potential differences in a closed loop is zero. Applications: circuit analysis, Wheatstone bridge and meter bridge.

## B6 Resistance

Series:  $R_s=4+6=10\ \Omega$ . Parallel:  $1/R_p=1/4+1/6=5/12$ ;  $R_p=12/5=2.4\ \Omega$ .

## B7 Photoelectric effect

Photoelectric effect is emission of electrons from a metal surface when light of suitable frequency falls on it. It proves particle nature of light. Applications: photocell and solar cell.

# Part C Solutions

## C1 Waves

Transverse wave: particle vibration perpendicular to direction of propagation. Example: light wave. Longitudinal wave: particle vibration parallel to direction of propagation. Example: sound in air. In one time period  $T$ , wave travels one wavelength  $\lambda$ . Hence  $v=\lambda/T=f\lambda$ . Numerical:  $v=340\text{ m/s}$ ,  $f=680\text{ Hz}$ .  $\lambda=v/f=340/680=0.5\text{ m}$ .

## C2 TIR and optical fiber

TIR occurs when light travels from denser to rarer medium and angle of incidence is greater than critical angle. In optical fiber, core has higher refractive index than cladding, so light undergoes repeated TIR and travels through core. Applications: telecommunication, internet, endoscopy and sensors. Malayalam: Fiber-ൽ light core-ൽ reflect ചെയ്ത് മുന്നോട്ട് പോകുന്നതിനാൽ signal long distance പോകാം.

## C3 Galvanometer conversion

Moving coil galvanometer detects small current by deflection of a coil in magnetic field. Ammeter conversion: connect low resistance shunt in parallel;  $S=IgG/(I-Ig)$ . Voltmeter conversion: connect high resistance in series;  $R=V/Ig-G$ . Malayalam: Ammeter ആക്കാൻ shunt parallel; voltmeter ആക്കാൻ high resistance series.



C4 Ammeter numerical

$G=99\ \Omega$ ,  $I_g=0.01\text{ A}$ ,  $I=1\text{ A}$ .  $S=I_gG/(I-I_g)=(0.01\times99)/(1-0.01)=0.99/0.99=1\ \Omega$ . Final answer: connect  $1\ \Omega$  shunt in parallel.

C5 Energy bands

In solids, electron energy levels form bands. Valence band and conduction band are separated by forbidden gap. Conductors have overlapping bands or no gap. Insulators have large gap. Semiconductors have small gap; doping changes their conductivity.

C6 LASER action

LASER works by stimulated emission. Pumping creates population inversion. An incident photon stimulates an excited atom to emit another identical photon. This produces coherent, monochromatic, directional and intense light. Applications: cutting, welding, optical communication, eye surgery and medical treatment.

MCQ Answer Key

1. B -  $v=f\lambda$ . 2. B - dioptre. 3. C - sum. 4. B - electrons. 5. C - coherent.

REFERENCES

Text / Reference books from syllabus

| Code | Book Title / Author                                                              |
|------|----------------------------------------------------------------------------------|
| T1   | Text Book of Physics for Class XI & XII, Part-I and Part-II, N.C.E.R.T., Delhi   |
| R2   | Applied Physics, Vol. I and Vol. II, TTTI Publications, Tata McGraw Hill, Delhi  |
| R3   | Concepts of Physics by H. C. Verma, Vol. I and II, Bharti Bhawan Ltd., New Delhi |
| R4   | Fundamentals of Physics, Halliday / Resnick / Walker, Wiley India Pvt. Ltd.      |
| R5   | Modern Approach to Applied Physics-I and II, A. S. Vasudeva, Modern Publishers   |

## Concept checklist

- ✓ I can define SHM, TIR, capacitance, photoelectric effect and population inversion.
- ✓ I can explain optical fiber, galvanometer, p-n junction and laser action.
- ✓ I can draw important diagrams.

## Numerical checklist

- ✓ I can use  $v=f\lambda$  and beat frequency.
- ✓ I can solve lens power and lens formula.
- ✓ I can solve resistance combinations and meter bridge.
- ✓ I can solve galvanometer conversion.
- ✓ I always write units.